DS Assignment 6A

Implement Bully and Ring algorithm for leader election.

```java
//BullyAlgo.java
import java.util.*;

public class BullyAlgo {
    static int n, coordinator;
    static boolean[] alive;

    static void election(int initiator) {
        if (initiator < 1 || initiator > n || !alive[initiator]) {
            System.out.println("Invalid initiator.");
            return;
        }

        System.out.println("Election started by Process " + initiator);
        coordinator = -1;

        for (int i = n; i >= 1; i--) {
            if (alive[i]) {
                coordinator = i;
                break;
            }
        }

        if (coordinator == -1)
            System.out.println("No coordinator. All processes are dead.");
        else
            System.out.println("New Coordinator: Process " + coordinator);
    }
```

```java
static void display() {
    System.out.print("Alive: ");
    for (int i = 1; i <= n; i++)
        if (alive[i])
            System.out.print(i + " ");

    System.out.println("\nCoordinator: " +
            (coordinator == -1 ? "None" : "Process " + coordinator));
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of processes: ");
    n = sc.nextInt();

    alive = new boolean[n + 1];
    for (int i = 1; i <= n; i++)
        alive[i] = true;

    coordinator = n;
    display();

    while (true) {
        System.out.println("\n1.Crash  2.Recover  3.Display  4.Exit");
        int ch = sc.nextInt();

        if (ch == 1) {
            System.out.print("Enter process to crash: ");
```

```java
    int p = sc.nextInt();

    if (p < 1 || p > n || !alive[p]) {
        System.out.println("Invalid process.");
        continue;
    }

    alive[p] = false;
    System.out.println("Process " + p + " crashed.");

    if (p == coordinator) {
        System.out.print("Enter initiator: ");
        int init = sc.nextInt();
        election(init);
    }

} else if (ch == 2) {
    System.out.print("Enter process to recover: ");
    int p = sc.nextInt();

    if (p < 1 || p > n || alive[p]) {
        System.out.println("Invalid process.");
        continue;
    }

    alive[p] = true;
    System.out.println("Process " + p + " recovered.");

    if (p > coordinator) {
        coordinator = p;
```

```
                System.out.println("Recovered process becomes coordinator.");
            }

        } else if (ch == 3) {
            display();

        } else if (ch == 4) {
            break;

        } else {
            System.out.println("Invalid choice.");
        }
    }

    sc.close();
  }
}
```

Output-

```
javac BullyAlgo.java
java BullyAlgo
Enter number of processes: 5
Alive: 1 2 3 4 5
Coordinator: Process 5

1.Crash  2.Recover  3.Display  4.Exit
1
Enter process to crash: 5
Process 5 crashed.
Enter initiator: 2
Election started by Process 2
```

New Coordinator: Process 4

1.Crash 2.Recover 3.Display 4.Exit

2

Enter process to recover: 5

Process 5 recovered.

Recovered process becomes coordinator.